

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

Chapter 6: We Distribute, Yet Things Multiply

Every formula and result of the chapter on one sheet

1	Distributive property	$a(b + c) = ab + ac$
2	Product of two brackets	$(a + m)(b + n) = ab + an + mb + mn$
3	Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$
4	Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$
5	Difference of squares	$(a + b)(a - b) = a^2 - b^2$
6	Sum of the two squares	$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
7	Difference of the two squares	$(a + b)^2 - (a - b)^2 = 4ab$
8	Śrīdharācārya's square	$a^2 = (a + b)(a - b) + b^2$
9	Cube of a sum	$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
10	Difference of cubes	$(a - b)(a^2 + ab + b^2) = a^3 - b^3$
11	Both numbers increased by 1	$(a + 1)(b + 1) = ab + a + b + 1$
12	Multiplying by 11	$N \times 11 = 10N + N$
13	Multiplying by 99	$N \times 99 = 100N - N$
14	Consecutive numbers	$n^2 - (n - 1)(n + 1) = 1$